DO NOW — RECALL

Lesson 087 · Algorithm Design and Testing Challenge

Name: _____ Class: _____ Date: _____

1. What is an invariant?

2. Why is sorting unnecessary for just finding one maximum?

3. What is nullable int?

4. Binary search for 8 in [2,4,6,8,10]. Give midpoint indexes and result.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A condition maintained while processing data.

2. Model answer

A single scan can find it in O(n).

3. Model answer

An int? that can be an integer or null.

4. Model answer

mid2=6, then mid3=8; found index3.

By the end, you can…

• Specify output when inputs are empty, and preserve correctness on all-negative arrays.

• Maintain invariant: best is the largest value among those inspected so far.

• Compare against alternative sorting-based solution: an O(n) scan avoids O(n log n) sort when only the maximum is needed.

Vocabulary — know these terms

maximum — Largest value in set.

invariant — Maintained fact during loop.

nullable — Type allowing absence/null.

complexity — Resource growth as input increases.

NEW LEARNING

Read, annotate and explain

Observe

Specify output when inputs are empty, and preserve correctness on all-negative arrays.

Explain

Maintain invariant: best is the largest value among those inspected so far.

Evaluate

Compare against alternative sorting-based solution: an O(n) scan avoids O(n log n) sort when only the maximum is needed.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace Maximum([7,3,10,4]).

Model response

The invariant tracks largest seen, not last examined.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace Maximum([-5,-2,-8]) showing all changes and final value. [6]

Question 2

Write algorithm for maximum of array, including empty input case. [7]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare O(n) maximum scan with sorting first and describe useful test coverage. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Initialising maximum to zero works for every integer array.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.